

Data Leverager Project Report

This document summarizes the data sources used, transformations applied, and challenges faced during the Power BI ETL and Power Query project.

1. Sources Used

- **Web source:** https://en.wikipedia.org/wiki/Social_media
- **Excel sales files :** Restaurant Operations and Sales Analysis 2023 ('Customer Details' , 'Food_Details' , 'Resturant_Details' , 'Order Data')
- **Employee data Excel file** (Employee_Datasets.xlsx)

2. Transformations Applied (name of file in which i have applied)

- Removed blank rows and duplicate records(All)
- Promoted first row as headers(All)
- Changed data types for cost, and dates(Employment Detail , Order Data)
- Applied Trim and Clean functions on text columns (Employ Profile)
- Converted text to uppercase (Employ Profile)
- Split address columns using delimiter(2 Customer Details)
- Created Salary Category using Conditional Column (Employment Detail)
- Added Index Columns [0-based] (Restaurant Detail)
- Extracted Day, Month, Year, and Quarter from dates(Order Data)
- Merged Employee Profile and Employee Details (Employee Profile)
- Appended all monthly sales files (Order Data)
- Performed Group By aggregation for sales analysis
- Enabled column quality and column profile features(All)
- Created parameter for dynamic folder path(All)

3. Challenges Faced and Solutions

Challenge	Solution
Null and blank values in data	Removed blank rows and filtered null values
Incorrect data types	Changed data types using Power Query
Extra spaces in text fields	Used Trim and Clean functions
Combining multiple files	Used Folder Source and Append Queries
Matching records between tables	Used Merge Queries with common columns
Automatic data updates	Used Refresh and Parameters

4. Conclusion

This project demonstrated the use of Power Query and Power BI for ETL operations including cleaning, transforming, merging, appending, and analyzing data from multiple sources.